GESTER
TOTAL TESTING SOLUTION



GT-C50
BEAN BAG SNAG TESTER

Operating manual

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Thank you for choosing the Gester Product. Please read this information guide carefully to be able to safely install, use and maintain your product

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COMPANY HISTORY

GESTER is a professional research, designing, manufacturing company, with the purpose of providing cost effective solutions for various industrial measuring applications, including: Textile, Fabric, Toy, Yarn, leather, Footwear, Furniture industries. Our company has grown to be a major international industrial testing equipment corporation with a long history, integrating development, production and distribution of testing equipments throughout China and International. All the products meet with: ISO, ASTM, AATCC, BS, EN, DIN, JIS and other requirable standards.

GESTER International has the factory in QUANZHOU of FUJIAN. To provide our customers with better overall solution laboratory, including laboratory design, planning, renovation and equipment selection, installation, training, maintenance, Comparative testing management system, such as one-stop authentication technology services.

SAFETY INSTRUCTIONS

Due to the potential hazards associated with any electrical instrument it is important that the user is familiar with the instructions covering the capabilities, and the operation of the instrument. The user should ensure that all reasonable safety precautions are followed and if in any doubt should seek professional advice before proceeding.

The instrument is designed for use by suitably trained, competent personnel in a controlled working environment and is intended for use as a universal wear tester only.

The instrument contains moving parts and while all reasonable steps have been taken to protect personnel associated with these moving parts, incorrect or mis-use of the instrument could result in injury.
Consideration should be given to the nature of these moving parts before setting up the instrument.

The instrument is intended to be used in a residential, commercial and light industrial environment as laid down in EN 50081-1.

WARNING

This unit contains moving parts and hazardous live voltages. Under no circumstance should the user try to prevent or restrict the movement of parts or gain access to the internal circuitry, either personally or with the aid of foreign bodies.
All ventilation slots must be kept clear.

PROVISION FOR LIFTING AND CARRYING

When unpacking or moving this unit extreme care is required, owing to its physical construction and weight.
It is recommended that accepted lifting and carrying procedures are employed and that personnel wear the appropriate protective equipment e.g. safety shoes.
If the unit is to be moved an appreciable distance/height it is recommended that it is moved via a suitable vehicle e.g. a fork lift truck

OPERATING ENVIRONMENT

This unit is intended to be used in a residential, commercial and light industrial environment as laid down in BSEN 50081-1 and BSEN 50082-1.

The following list gives examples of locations in which the instrument might be located; workshops, laboratories and service centres. Locations which are considered to be commercial or light industrial.

CLIMATIC ENVIRONMENT

The unit is intended to operate within the following conditions

i) temp 5-55 deg Celsius

ii) humidity 30-80% RH

iii) Altitude up to2000m above sea level.

And is intended to be stored in a temp range of –25 - +25 deg Celsius.

ELECTRICAL INFORMATION

This unit complies with BSEN 61010-1 1993 safety Requirements for Electrical Equipment for Measurement, Control and Laboratory Use.

INSTALLATION CATEGORY AND POLLUTION DEGREE

Installation category III

Pollution Degree 2

SECTION 1 INTRODUCTIONS

Application

To determinate the performace of knitted fabric snagging resistance.

Theory

Put one bead pillow into one knitted sample, and put the knitted sample into one expansion cylinder which has needle on the inner wall row by row to rotate a certain number of turns. Take the tested sample to evaluate the snagging resistance grade.

Standards

ASTM D5362 JIS L1058

Structure

1. Structure:



1.Revolving Drum
2.Nail Frame
3.Acrylic cover
4.Control Panel

2. Function:

- Revolving Drum:2 test rooms
- Nail Frame : do the effect to the sample for test.
- Control Panel: include control instrument、indicate light and control switch button.
- Sampling Tempalte: used to cut samples in right size.

Control Panel

1.Control Panel



1. Counter 2. Start 3. Stop

2. Control Panel Introduction.

- COUNTER.: Pre-set the test times and display the current operation number.
- START: Press START, the revolving drum begins to test.
- STOP: Press STOP, the revolving drum stops.

Specification

Item	Standards
Rotate Speed	20±3 r/min
Revolving Cylinder	Inner diameter:200mm，depth:145mm
Nail Frame	8 Nail bar / Nail Frame
Nail Bar	Outside diameter：8mm，length：127mm，9 Nail / Nail Bar
Nail	Protruding Length:10mm,Distance:12mm,Angle with vertical direction :30±5°
Smaple size	215mmx115mm
Volume (WxDxH)	47x38x38cm
Weight(About)	≈38Kg
Power	1∮ AC 220V 50Hz 3A

Accessories

- Smapling Template：215mmx115mm
- Bead pillow: 2，450±10g/each one

- Rating Light Box(In addition to buy)
- Rating figure card (In addition to buy)
- Sewing Tool (autonomous)

SECTION 2 INSTALLATION

Power condition

Please install correct power supply according to the marking on the tester brand.

Working Environment Requirement

1. Room Temperature Condition.

2.This machine should put on the platform that is level and steady.

NOTE



The input voltage error range is within ±10% , ensure the tester grounded correctly, to prevent the harm of leakage.

SECTION 3 SAMPLE PREPARATION

Take samples

- ✧ Take Samples—— a reel of fabric as the sample unit one time, random take one reel from useable raw material,or take the samples that agreed by buyer and seller.
- ✧ Take sample in the lab——The original sample volume should be 1m after abandon the outer part.
- ✧ Sample——Test three pieces of sample.

Sample prepartion

- ✧ Laundering—— put the sample with the same material and quality (such as the same manufacturer,same yarn,same work and same prophase maintenance)、the whole weight 3.5kg or 81b in to washing machine, or take one group congeneric that after desizing but not soften into the washing machine. Select the normal working procedure,warm water,use the cleaner that mentioned in AATCC standard (see the D3136 or AATCC 135 test method). Working for one work period without

softener.Take the washed cloth in to dehydrator.Select the normal working procedure, medium temperature, dehydrator work for 20mins or work to when the cloth is dry.Don't use softener in the hydrator, but don't excessive jilt dry cloth.

- ✧ Use the sample template to cut 6 samples of 215mm×115mm from determinand fabric.
- ✧ Overlap every sample short side ,the inner side of overlap sample is the surface that is use to Bean Bag Snag test.
- ✧ The new sample size is 107mmX115mm after folding, use machine or hand to sew the sample two sides of length 107mm to be a sample bag, Suture density is about 0.4 sitch/mm(10 sitch/in)
- ✧ Turn the sew up sample inside out, put the bean pillow into the sample bag through the opening,then sew the opening up, the suture density is 0.4 sitch/mm(10 sitch/in), the distance from suture to edge is more 6.5mm, trim all the suture edge, make the sample out the suture is less than 3mm.

【Note】

Put the under test fabric long edge parallel with template long edge,the Bean Bag snag tester shows random rotation movement will not make the fabric to be obvious directive effect, Because no obvious directive effect ,so the lengthways and transverse sample is not all need to test.

SECTION 4 OPERATION

Here are some item you should confirm before your operation.

- ✧ Ensure the power is comply with the requirement.
- ✧ Take out the nail frame, check the nail bar and rotate cylinder is tidy or not. If there is need cleaning, please wear the protective gloves.
- ✧ Check the nail is damage or losseness or not.

Put the nail frame into rotate cylinder, make the nail front and rotate cylinder front parallel and level.

Unscrew the button on the nail frame front, take down the cover plate, put the prepared sample into rotate cyclinder nail frame bottom, seam place upside, put one sample into

every cyclinder, cover the cover plate and screw the button tight.

Begin to test:

- ✧ Power on, set the test time (100 time)and reset the count to zero.
- ✧ Press START to begin test, the machine will stop automatic when come to the setting time.
- ✧ Unscrew the button on the nail frame front ,take down the cover plate, take out the sample.
- ✧ Pull down the suture on the sample .

Danger



Please ensure wear the protective gloves and turn off machine power when you pick and place sample or check the nail frame, to prevent the nail from hurting the operator.

SECTION 5 EVALUATION AND REPORT

You can choose method A (Figure card method) or method B (count method) to rate.

- ✧ Method A(Figure card method):compare the sample with the pilling figure card on the rating light box to obtain the rate.
- ✧ Method B (Point method):
  - ➢ Observe under the white cold fluorescent tube(color temperature 4100~4500K),lay the sample flat on the observation desk,collapse edge is near the chest, the sample centre is 300mm distance from the eyes, angle of view keep 0~45 degree(the angle with vertical direction). Use the point system to count the amount of visible points on the sample two surface, and take the average value of two surface result, correct to 0.5 unit:
  - ➢ Snag protuberance but no warping,called 1 point
  - ➢ Snag protuberance and warping,called 2 point.
  - ➢ Snag warping but no protuberance ,called 3 point.
  - ➢ Rate reference to the level evaluation method as below:

Rating Total Points（Two surface） Clarify phenomenon

5 ----------0 to 2 ----------No or negligible snag

4 ----------3 to 10 ----------Slight snag

3 ----------11 to 19 ----------Moderate snag

2 ----------20 to 39 ----------Severe snag

1 ----------40 or more ----------Very severe snag

✧ According to the sampling unit of each sample, calculate the average value from the three sample data, correct to 1.0 scale base.

【Note】



■ When you take the Method B(Point Method)， protuberance and warping must be obvious. That means when you observe snag, it's difficult to judge the protuberance 、warping or include both ,these situation are invalid.

■ All the record above is for reference, please refer to the relative standard for details.

✧ Test 6 samples ,check whether there is color contrast, Long distorted(the length is more than 15mm) or long bumps(the length is more than 4mm) situation or not. If there are at least 2 samples appear this situation in 3 samples, please record in test report.

✧ Report should include the items below

➢ The method to prepare sample ,include scald application.

➢ The rotate cyclider quantity when you test each sample .

➢ Use other method when you do evaluation.

➢ If you choose option A, use which grade standard.

✧ If you choose option B, introduce point system.

✧ According to the each unit sampling unit, record the sample average grade balue and some appearance change.

【NOTE】



If you want to stop halfway,press STOP , it will stop action.

All the record above is for reference, please refer to the relative standard for details.

SECTION 6 CALIBRATION PROCEDURE

- Calibration Item: Speed.
- Calibration Appliance: Cronometro.
- Calibration Period: 1 year.
- Calibration Step:
  - Speed calibration method
  - Turn on Power on the machine ,and set the test times more than 20 times.
  - Press Start to start the machine ,and cronometro begin to count at the same time.
  - When the cronometro come to 1 mins, stop the cronometro, and press STOP, themachine stop, check whether the times display on the cronometro is comply with the rotate speed or not.

SECTION 7 MAINTAIN PROCEDURE

- Please clean the machine surface before and after test every time.
- Please add lube to machine moving parts regularly.
- The machine long time no use ,please pull out the power plug.

SECTION 8 TROUBEL SHOOTING

Fault Situation	Possible Cause	Solution
• Turn on the power switch , the indicate light is not on..	• No power supply • The fuse burn out	• Please ask for the electrician to check and recover the power supply. • Change the fuse with the same capacity.
• There is big abnormal noise when the machine is working .	• Motor Fault • Reduction gear Fault • Moving part bearing Fault.	• Maintain or change motor with the same standard. • Change reduction gear with the same standard. • Change the bearing with the same standard.
• Press Start ,the machine is not working.	• Check the whether the “count/time” reach the set value or not.	• Set “count/time” controller set value again,return to zero ,press Start to begin.
• The fuse burn out frequently, out of use.	• The input voltage is abnormal • Motor abnormal	• Please ask the electrician to check and recover the normal power supply. . • Motor damage, maintain or change motor .
• The machine works well,but the counter cannot count.	• The inductor induction is abnormal. • The inductor fault. • The count fault.	• Adjust proximity switch induction screw distance to a right place. • Change inductor. • Change count.

【Note】



If there are problems you can’t deal with ,please contact the sevice department in our company.

SERIAL NUMBER_____

DATE _____

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